: Grounding Adaptable and Scalable Generation of Football Tactics

TacticGen reframes football analytics from predicting what may happen to generating what should happen. It models coordinated player and ball movements with a multi-agent diffusion transformer, then guides generation toward tactical objectives expressed as rules, natural language, or learned value functions.

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Paper ↗ Code ↗ Partner ↗

3.3M+
annotated events for training

100M
real tracking frames

1432
matches from 2018-2025

80%
of experts favored TacticGen

HIGHLIGHTS

Why this paper matters

TacticGen is positioned as a **foundation-oriented model** for football tactical design: **accurate enough** to model realistic movement, **adaptable enough** to respond to different objectives, and **scalable enough** to improve with larger models and more data.

Top

01

From prediction to generation

The paper reframes football analytics from **forecasting likely play evolution** to **generating coordinated tactical alternatives**.

02

Multi-agent football modeling

TacticGen jointly models **the ball and 22 players**, explicitly handling **cooperative and competitive interactions**.

03

Zero-shot flexible tactical guidance

During inference, a single trained generator can be steered by **rule functions**, **natural language prompts**, or **learned value models**.

04

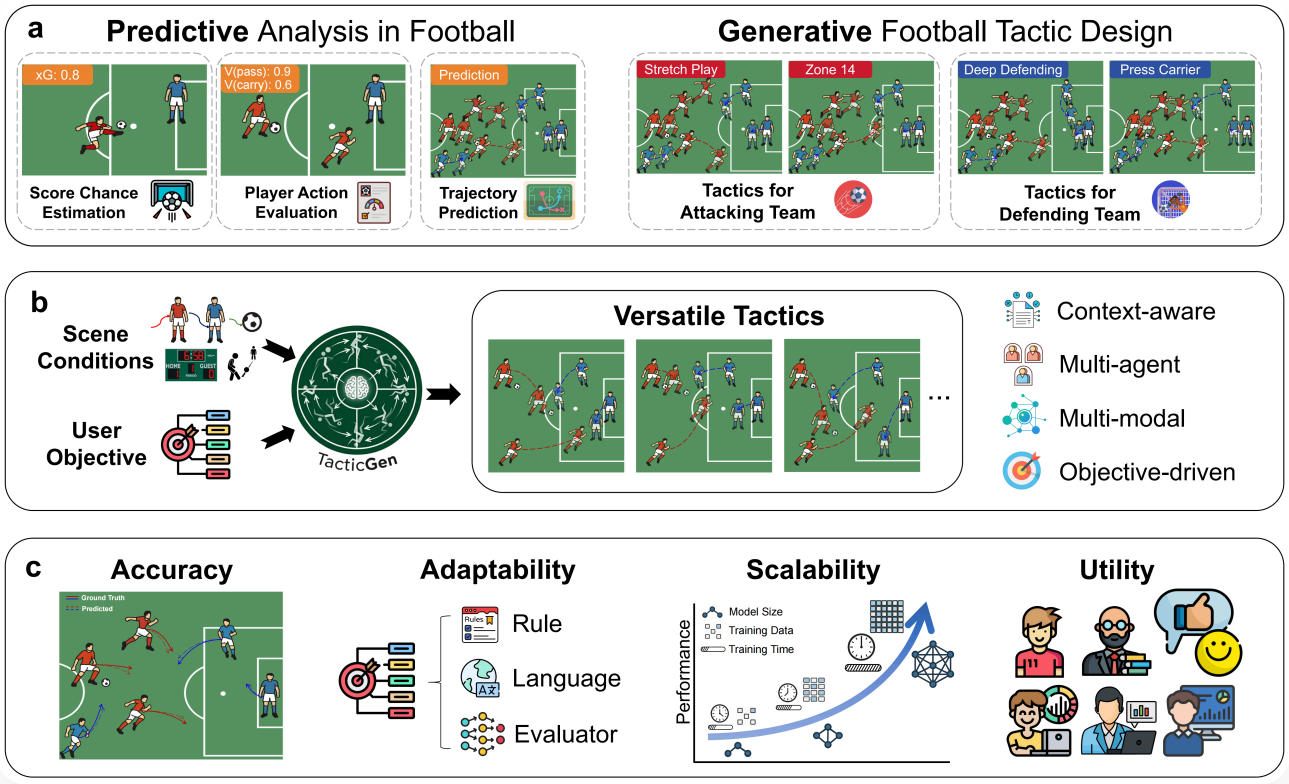
Validated with practitioners

Case studies with football experts show generated tactics are **hard to distinguish from real play** and **tactically valuable**.

MOTIVATION

Why football tactics should be generated, not only predicted

This figure sets up the problem shift behind TacticGen: instead of only forecasting how play may unfold, the system should also produce coordinated tactical alternatives that align with a coach's intent.



Motivation Figure

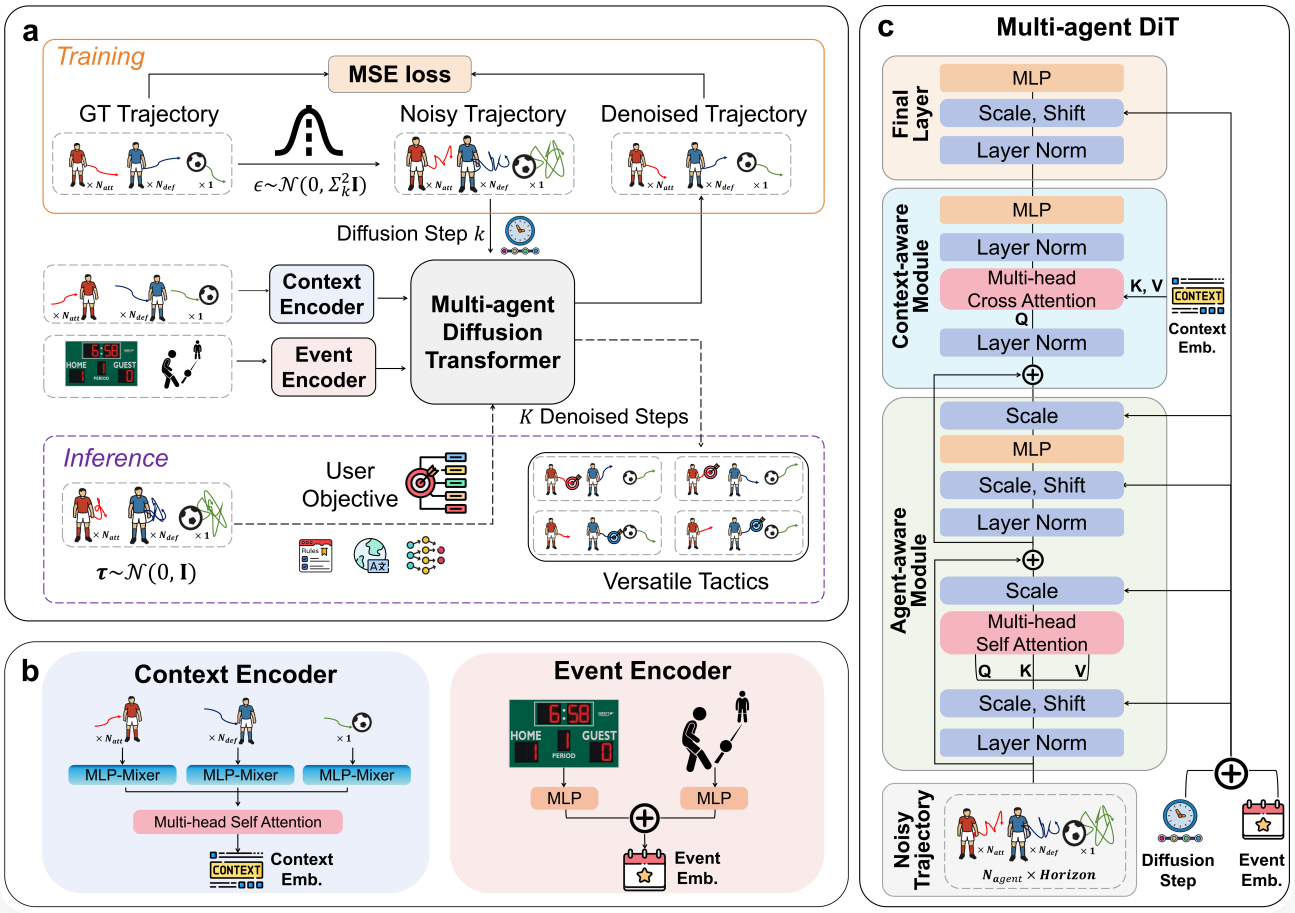
From descriptive football analytics to tactical decision support

The motivation figure introduces the paper's core question: how to move from passive trajectory prediction toward controllable generation that can assist tactical planning.

FRAMEWORK

The TacticGen pipeline

After the method overview, this figure gives a cleaner system-level view of how context encoding, diffusion generation, and tactical guidance fit together inside the complete framework.



Framework Figure

Conditioning, generation, and guidance in one model stack

The framework figure ties the paper together by showing how event context, player-ball history, and guidance signals are fused into a unified generation pipeline for tactical control.

METHOD

A controllable diffusion framework for football tactics

TacticGen builds on diffusion transformers and adapts them to football with explicit agent interaction modeling, context-aware conditioning, and guidance at sampling time.

ARCHITECTURE

The core generation pipeline

These three components define the base system: a multi-agent denoising backbone, a context encoder that summarizes match state, and an inference-time objective module that makes the generator steerable.

BACKBONE

DIT CORE

Multi-agent Diffusion Transformer

TacticGen extends DiT into a multi-agent setting so that noisy future trajectories can be denoised while preserving team-level coordination and opponent response.

Joint denoising across players and ball keeps team structure coherent.

CONTEXT

TEMPORAL FUSION

Self-attentive context encoding

Past trajectories, ball history, event type, goal difference, and timestamps are fused into context embeddings through MLP-based encoding, self-attention, and adaptive conditioning.

Rich match context is compressed into controllable conditioning signals.

GUIDANCE

INFERENCE CONTROL

Objective-driven generation

Classifier guidance steers samples at inference time, letting one pretrained generator adapt to tactical goals without retraining.

One generator, multiple tactical objectives, no extra training pass required.

On top of the shared architecture, TacticGen exposes multiple control interfaces so the same pretrained model can respond to different tactical design goals.

GUIDANCE

Three ways to steer tactical generation

The guidance layer supports explicit football rules, coach-friendly language prompts, and learned value signals, covering both interpretable control and performance-oriented optimization.

01



Rule-based guidance

Differentiable tactical rules encode concepts like support width, zone occupation, or pitch control value.

Best for explicit tactical priors and interpretable control.

02

Language-based guidance



Natural language instructions are turned into executable guidance functions, enabling high-level tactical prompting.

Best for coach-friendly prompts and rapid what-if iteration.

03

Value-based guidance



A learned value model injects long-term utility into generation, guiding trajectories toward strategically stronger outcomes.

Best for optimizing downstream tactical quality over longer horizons.

The same architecture is deployed in two practical operating modes, letting TacticGen cover both realistic trajectory prediction and more deliberate tactical design workflows.

OPERATING MODES

Two practical operating modes

Both modes share the same football-native generator, but differ in how much future ball information is available at inference time and, as a result, what kind of coaching workflow they support best.

TACTICGEN-P

PREDICTIVE

Short-context trajectory prediction

Predicts future player and ball trajectories from a short observed context window, making it a natural fit for realistic generation.

- Uses recent match context without assuming the future ball path is known.
- Best suited to realistic future trajectory prediction and model benchmarking.

TACTICGEN-C

CONDITIONAL

Ball-conditioned tactical generation

Conditions on a full ball trajectory so the model can coordinate player movement around an anticipated ball path.

- Provides stronger control when analysts or coaches already have a target ball path in mind.
- Best suited to tactical planning, design exploration, and controlled what-if generation.

TacticGen project page.